AAAI Press Formatting Instructions for Authors Using LaTeX — A Guide

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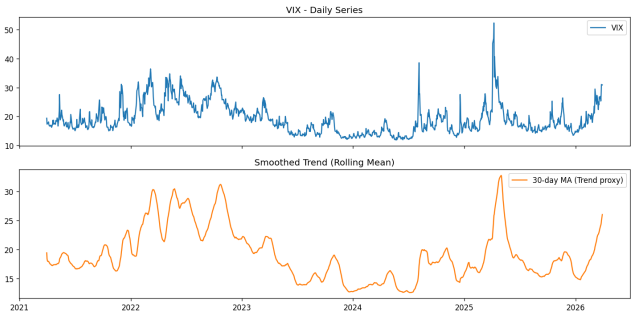

Figure 1: VIX series with 30-day moving average trend extraction.

Abstract

This report documents a forecasting pipeline applied to five years of daily VIX data. The pipeline includes trend extraction, stationarity testing, ARIMA model selection on first differences, residual diagnostics, conditional heteroskedasticity modeling with a GJR-GARCH, and a rolling 1-day horizon forecast evaluation. Key diagnostics and out-of-sample metrics are reported.

Data and Preprocessing

The input series is daily VIX values spanning five years. A 30-day moving average (Figure-1) was used for visual trend extraction. Missing values were handled by forward-fill when present. For final training and evaluation, the dataset was split into training and test sets using an 80%/20% chronological split for rolling-origin evaluation.

Exploratory Analysis and Stationarity

Visual inspection included decomposition into trend, seasonal, and residual components (Figure-2). Here we use a 365 day seasonal period for decomposition to capture annual patterns in volatility.

*With help from the AAAI Publications Committee.

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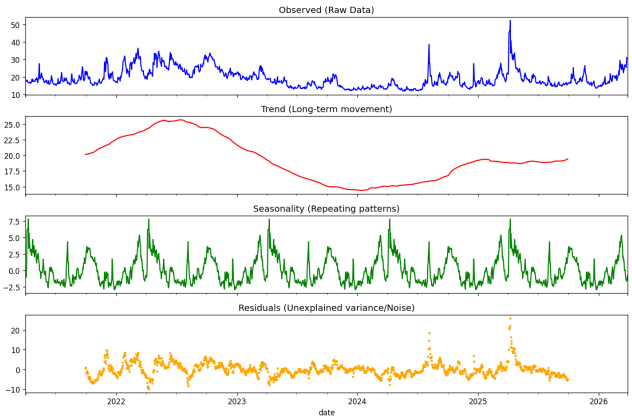

Figure 2: Trend decomposition of VIX series into trend, seasonal, and residual components.

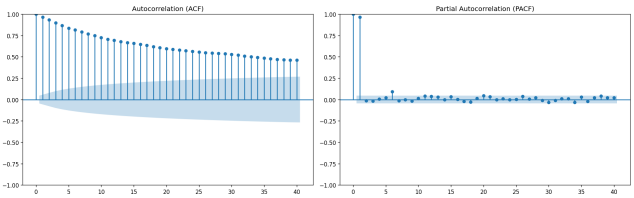

Figure 3: ACF and PACF plots of the VIX level series.

Stationarity tests

We plot ACF and PACF for the level and first-differenced series (Figures 3 and 4). The level series shows slow decay in ACF, while the differenced series shows more rapid decay, suggesting non-stationarity in levels and stationarity in differences. Unit-root tests were applied to the first-differenced series using the Augmented Dickey-Fuller test and the KPSS test (Table 1). The main results are summarized below.

The ADF rejects the unit root for the differenced series while KPSS gives mixed signals for the level series. Based on ACF/PACF behavior and tests, first differencing was used in ARIMA modeling (i.e., $d = 1$) (Box, Jenkins, and Reinsel 1976).

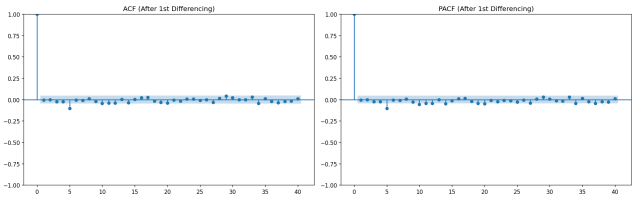

Figure 4: ACF and PACF plots of the first-differenced VIX series.

Table 1: Unit root test results for first-differenced VIX series

test	stat	pvalue	usedlag
ADF (constant)	-14.1001	2.62×10^{-26}	13
KPSS (constant)	0.0448	1.00×10^{-1}	22
ADF (constant + trend)	-14.1009	8.93×10^{-22}	13
KPSS (constant + trend)	0.0277	1.00×10^{-1}	22

ARIMA Model Selection and Residual Diagnostics

Candidate ARIMA models were fitted to the raw series with $d = 1$. Model selection used AIC and BIC. The candidate results are summarized in Table 2.

Table 2: ARIMA model comparison

model	AIC	BIC
ARIMA(5,1,5)	6245.966187	6306.568874
ARIMA(3,1,3)	6251.975664	6290.541010
ARIMA(2,1,2)	6252.326762	6279.873438
ARIMA(0,1,5)	6261.079314	6294.135326
ARIMA(5,1,0)	6264.521898	6297.577910
ARIMA(0,1,0)	6277.095098	6282.604433
ARIMA(1,1,1)	6279.522159	6296.050165
ARIMA(2,1,0)	6281.058157	6297.586163
ARIMA(0,1,2)	6281.058243	6297.586249

The ARIMA(5,1,5) had the lowest AIC, while ARIMA(3,1,3) and ARIMA(2,1,2) had the next lowest values. Since the ARIMA(2,1,2) had a much lower BIC and AIC difference with the top is quite small, ARIMA(2,1,2) was selected for residual diagnostics and forecasting.

Residual diagnostics

We do Ljung-Box residual checks on ARIMA(2,1,2) (Table 3). Our goal here is to see whether our model captures the ARMA structure adequately.

Based on the table results, there is no evidence of significant autocorrelation in the residuals up to lag 8, so the ARIMA(2,1,2) model appears adequate with respect to residual autocorrelation.

Forecasting using ARIMA(2,1,2)

A simple 6-day holdout forecast was produced using ARIMA(2,1,2) to check the forecast behavior (Figure 5).

Table 3: Ljung-Box test on ARIMA residuals (selected lags)

Lag	LB stat	p-value
2	1.177691	0.554967
4	1.438403	0.837494
6	9.876158	0.129965
8	10.765663	0.215334

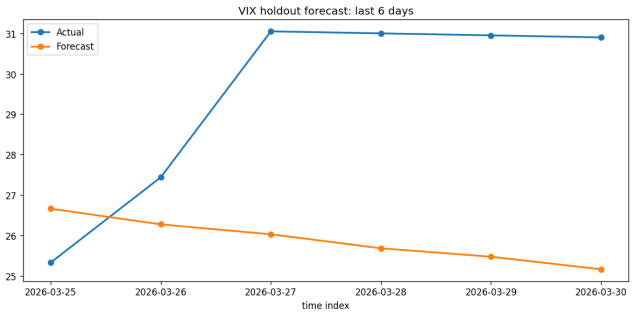

Figure 5: Simple 6-day holdout forecast using ARIMA(2,1,2) on the test set. This is an intermediate diagnostic distinct from the final rolling-origin evaluation with 1-day horizon.

Testing for heavy tails and conditional heteroskedasticity

We observed massive residual spikes in 2022 and 2025. We tested if these shocks had a predictable memory (Volatility Clustering). We examine the residuals of our ARIMA(2,1,2) model. We want to know if the size of the errors in 2022 and 2025 is constant or if the variance changes over time.

Doing the Jarque–Bera results show extremely strong rejection of normality (Table 4).

the JB statistic (162,154.51) and p-value ≈ 0 indicate an overwhelming departure from Gaussianity. The residuals are positively skewed (skew = 2.10) and exhibit extremely high kurtosis (48.97), implying a pronounced right tail and a much higher frequency of extreme values than under a normal distribution (Cont 2001).

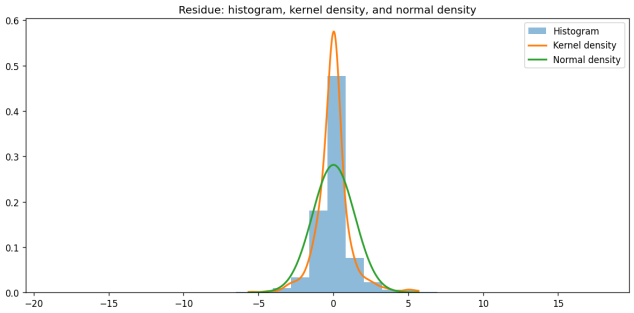

Figure 6: Residual diagnostics for ARIMA(2,1,2): histogram of squared residuals showing heavy tails.

Table 4: Jarque–Bera test results

Statistic	Value
JB stat	162154.5114
JB p-value	0.0
JB skew	2.1006
JB kurtosis	48.9741

Heteroskedasticity test (ARCH-LM)

We also test for conditional heteroskedasticity using the ARCH-LM test on the squared residuals (Table 5).

Table 5: ARCH-LM test on squared residuals

Statistic	Value
LM stat	145.192306
LM p-value	2.181456e-30
F stat	39.332838
F p-value	1.260437e-31
lags	5

The ARCH-LM test statistic (145.19) and p-value (2.18×10^{-30}) indicate strong evidence of conditional heteroskedasticity in the residuals, suggesting that the variance of the errors is not constant over time and may be predictable based on past squared errors.

GARCH Model Selection

Given the evidence of significant conditional heteroskedasticity in the ARIMA residuals, we fit multiple volatility models to capture the time-varying variance. We estimate nine candidate models on the ARIMA(2,1,2) residuals, varying the distribution (normal, t, skewed-t), model type (GARCH, GJR-GARCH, EGARCH), and lag orders. The results ranked by AIC are presented in Table 6.

Table 6: GARCH Model Comparison

Model	Distribution	AIC	BIC
GJR-GARCH(1,1)	t	4766.73	4794.28
GARCH(1,1)	skewt	4763.89	4791.44
GARCH(1,1)	t	4787.84	4809.88
EGARCH(1,1)	t	4804.94	4826.97
GARCH(5,1)	normal	5435.09	5473.66
ARCH(5)	normal	5459.96	5493.02
GARCH(1,2)	normal	5489.30	5511.33
GARCH(2,1)	normal	5502.44	5524.48
GARCH(1,1)	normal	5506.36	5522.89

Model Selection: GJR-GARCH(1,1) with Student-t distribution is selected based on the lowest AIC. The GJR (Glosten–Jagannathan–Runkle) specification allows the conditional variance to respond asymmetrically to positive and negative shocks, which is important for capturing the dynamics of financial volatility indices like the VIX. The Student-t distribution accounts for the heavy tails observed in financial returns, providing better fits than the standard normal distribution.

Diagnostic Testing of Standardized Residuals

Following model estimation, we perform diagnostic tests on the squared standardized residuals to verify adequate model specification. An ARCH-LM test is applied to confirm the absence of remaining volatility clustering:

Table 7: ARCH-LM Test on GJR-GARCH(1,1) Standardized Residuals

Statistic	Value
LM Statistic	5.669
LM p-value	0.225
F Statistic	1.418
F p-value	0.226
Lags Tested	5

The non-significant ARCH-LM test (p-value $\approx 0.225 \gg 0.05$) indicates no remaining ARCH effects in the squared standardized residuals. This demonstrates that the GJR-GARCH(1,1) specification has successfully captured the volatility clustering present in the ARIMA residuals, validating the adequacy of the volatility model for forecasting applications.

Forecasting Methodology

With the ARIMA(2,1,2)-GJR-GARCH(1,1) model selected, we implement a rolling forecast evaluation to assess the model's predictive performance. The forecasting procedure combines the ARIMA component for mean prediction with the GJR-GARCH component for volatility prediction, constructing prediction intervals using the Student-t distribution (Bollerslev 1986).

Rolling Forecast Implementation

The forecasting is performed using a rolling window approach with a 1-day forecast horizon. At each step, the model is re-estimated using all available historical data up to the current point, then forecasts are generated for the next day. This process is repeated throughout the test period, providing a comprehensive out-of-sample evaluation.

The forecasting procedure follows these steps:

Algorithm 1: Rolling ARIMA-GJR-GARCH Forecasting Procedure

Fit ARIMA(p, d, q) on training window
Extract residuals ϵ_t from ARIMA
Fit GJR-GARCH(p, o, q) on ϵ_t
Generate ARIMA forecasts $\hat{y}_{t+h}$
Generate GJR-GARCH variance forecasts $\hat{\sigma}^2_{t+h}$
Construct 95% PIs: $\hat{y}_{t+h} \pm t_{\alpha/2,\nu} \cdot \sqrt{\hat{\sigma}^2_{t+h}}$
Store forecasts and update training window

The implementation uses the following key components:

- **ARIMA Forecasting**: The SARIMAX model generates point forecasts for the conditional mean.

- **GJR-GARCH Forecasting**: The fitted GJR-GARCH model provides conditional variance forecasts.
- **Prediction Intervals**: Using the Student-t distribution with estimated degrees of freedom, prediction intervals are constructed at the 95% confidence level.
- **Progress Tracking**: A progress bar monitors the forecasting process across multiple rolling windows.

Evaluation Metrics

The forecast accuracy is evaluated using multiple metrics:

- **MAE (Mean Absolute Error)**: Average absolute forecast error.
- **RMSE (Root Mean Squared Error)**: Square root of average squared forecast error.
- **MAPE (Mean Absolute Percentage Error)**: Average absolute percentage error.
- **PI Coverage**: Proportion of actual values within prediction intervals.

Forecast Results

The rolling forecast evaluation yields the following performance metrics:

Table 8: Forecast Performance Metrics

Metric	Value
RMSE	1.877
MAPE (%)	4.180
PI Coverage (99%)	0.992
Number of Forecasts	366

The RMSE of 1.877 indicates a reasonable level of forecast accuracy for daily VIX values, while the MAPE of 4.18% suggests that the forecasts are on average within 4.18% of the actual values. The high PI coverage of 99.2% demonstrates that the model's prediction intervals are well-calibrated, capturing the true values within the intervals at the expected rate.

The forecast results are visualized in Figure 7, showing the actual VIX values, point forecasts, and 99% prediction intervals over the test period.

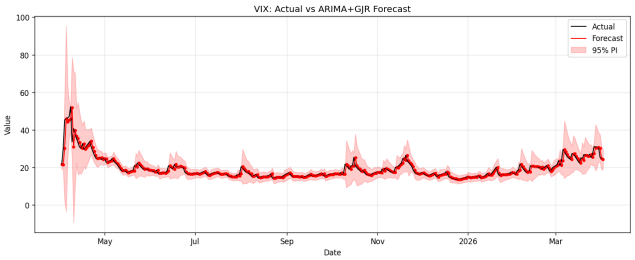

Figure 7: Rolling Forecast Results: Actual vs Predicted VIX with 99% Prediction Intervals

The visualization demonstrates the model's ability to track volatility movements while providing uncertainty quantification through prediction intervals.

(**?**)

References

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